D02 · S&OP; / IBP Governance

04 · Master data governance for planning

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1 WHAT IT IS

Master data governance for planning establishes ownership, definitions, validation, and a maintenance cadence for the data the planning engine consumes — lead times, BOMs, product and location hierarchies, safety-stock and order parameters — so the plan is built on data that is current, consistent, and trusted rather than on defaults set at go-live and never revisited.

2 WHY IT MATTERS

Redman 1998 **(Redman, 1998 — peer-reviewed)** documented that poor data quality imposes large, mostly hidden costs on the enterprise — rework, wrong decisions, and lost trust — and that the damage compounds the further downstream the bad data travels. In planning, master data is the most leveraged input: a wrong lead time, an outdated BOM, or a stale safety-stock parameter is silently applied to every run, every SKU, every cycle. The sophistication of the algorithm is irrelevant when the inputs are wrong — **a planning engine fed stale lead times produces confident, precise, and wrong plans**, and the organization trusts them precisely because they look rigorous.

3 HOW IT IS DONE

1. **Inventory the planning-critical master data.** List the fields the engine actually consumes — lead times, BOMs, hierarchies, lot sizes, safety-stock and service-level parameters — and rank them by impact on the plan. Govern the high-leverage few, not every field equally.
2. **Assign a named owner per data domain.** Every critical field has one accountable owner — procurement owns lead times, engineering owns BOMs. Data with no owner is data no one maintains; ownership is the foundation of governance, not the documentation about it.
3. **Define each field and its valid range.** Write down what "lead time" means (order-to-receipt? to-dock? to-available?) and the plausible bounds. Most master-data corruption is not malice but ambiguity — two functions populating the same field with two definitions.
4. **Validate against reality, not against the last value.** Compare planned lead times to actual receipts, BOMs to as-built, parameters to realized service. The check is whether the data matches what happened, not whether it matches what was entered before.
5. **Set a maintenance cadence and a change workflow.** Critical parameters are reviewed on a schedule, and changes flow through a controlled workflow with approval and an audit trail — not silent edits that no one can later explain.
6. **Monitor data-quality KPIs.** Track the share of lead times validated in the last N months, BOM accuracy, and parameter staleness. Master data degrades continuously; without a standing quality metric the rot is invisible until the plan fails.

> **WORKED EXAMPLE — ILLUSTRATIVE**
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> A manufacturer ran a sophisticated APS on master data last cleansed at implementation — **baseline: 38% of lead times more than two years old, BOM accuracy ~82%, planners routinely overriding the system because they did not trust it**. It assigned data owners, validated lead times against actual receipts, and set a quarterly review. Results: **validated lead times 62% -> 94%, BOM accuracy 82% -> 96%, planner override rate down by half** — and forecast and supply plans the team finally trusted, because the inputs were finally true.

IN PRACTICE

Data definitions and a single dictionary

Maintain one authoritative data dictionary for planning fields, with definition, owner, valid range, and source of truth. Ambiguity is the root cause of most inconsistency; a single dictionary removes the "we each meant something different" failure that no validation rule can catch.

Scheduled parameter review

Review high-leverage parameters — lead times, safety stocks, lot sizes — on a fixed cadence against realized performance. Parameters set once at go-live silently decay as suppliers, volumes, and networks change; the scheduled review is what keeps them tethered to reality.

Controlled change workflow

Route changes to critical master data through a workflow with approval, validation, and an audit trail. Uncontrolled edits — a planner quietly changing a lead time to fix one order — are how master data drifts out of trust one well-meaning edit at a time.

Data-quality KPIs in the S&OP scorecard

Put master-data quality metrics — validated-lead-time share, BOM accuracy, parameter staleness — on the S&OP scorecard alongside forecast accuracy. Making data quality visible at the leadership level is what funds and sustains the unglamorous maintenance work.

WHAT YOU WOULD USE

MASTER DATA GOVERNANCE / MDM

SAP Master Data Governance / Informatica MDM — enterprise tier-1 — Enterprise MDM with workflow, validation rules, stewardship, and audit trails across material, supplier, and location data. **(SAP, Informatica — vendor)** The choice when master data spans many systems and needs a governed single source feeding the planning engine.

Stibo / Reltio — MDM platforms — Capable multidomain MDM platforms with strong data-quality and stewardship tooling. **(Stibo, Reltio — vendor)** A strong fit where product and supplier data governance is the priority and the planning engine is one of several consumers.

Data-quality / observability specialists — Dedicated data-quality and monitoring tools that profile, validate, and alert on master-data drift. **(specialized vendors — vendor)** The vertical option when the need is continuous quality monitoring rather than full MDM.

For operations that don't need a platform — A data dictionary, named owners, a scheduled validation of lead times and BOMs against actuals, and a simple change log. Governance is a discipline before it is a platform — most of the trust is recovered by ownership and validation, not by buying MDM.

4 SO WHAT

> **Operations that govern planning master data feed the engine inputs the planners trust; operations that set parameters at go-live and walk away feed it confident garbage.** Master data is the least glamorous and most leveraged investment in planning — every algorithm, forecast, and optimization runs on top of it. The planner override rate is the clearest symptom: people stop trusting a system whose inputs they know are stale, and once trust is gone, the platform is just expensive theater.

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